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An Explainable XGBoost-Based Intrusion Detection Framework for Cloud Security Using SHAP Analysis

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ABSTRACT:

With the proliferation of cloud-based services and the volume of data being stored and managed on this platform, businesses are exposed to new risks from cyber threats like Distributed Denial-of-Service (DDoS) attacks, exploits, reconnaissance, malware communication and unauthorized access attempts. While conventional IDSs are less effective in detecting new attack patterns and typically are opaque in their decision making, they can prevent the occurrence of attacks.Traditional IDSs are generally opaque in their decision-making, and may be less effective in detecting new attack patterns, but they can be used to stop attacks from happening. To overcome such issues, this paper introduces an Explainable XGBoost Based Intrusion Detection Framework for Cloud Security Using SHAP Analysis. The methodology takes advantage of the dataset UNSW-NB15, which preprocesses network traffic records, encodes them and classifies them with a machine learning algorithm: XGBoost. Aimed at improving model interpretability, a new ingredient is added: SHAP (SHapley Additive exPlanations) is incorporated to explain the role of individual features from the network to the classification results. The results of the experimental evaluation indicate that the proposed framework provides an overall accuracy of 95.13% and high values of precision, recall, and F1-score, which indicate the effectiveness of the framework in differentiating between malicious traffic and normal network behavior. The findings also indicate that explainable artificial intelligence can offer valuable insights into the decisions made for detecting attacks, which can boost trust and transparency in cloud security systems. Overall, the study demonstrates that XGBoost with SHAP is a promising and explainable approach for intrusion detection in today's cloud computing landscape.

1. Introduction

Cloud computing has become one of the most revolutionary technologies in the computer age providing scalable, flexible, and cost-effective resources to individuals, enterprises, and governments. Cloud services have allowed companies to scale up, store massive amounts of data, deploy apps with ease and access computing resources when they need them, without the extensive investment in infrastructure. As organizations move to cloud environments, however, the cybercriminal landscape has grown as a result, and cyber security is a major issue for any organization in the world today [13] and [15]. Distributed Denial-of-Service (DDoS) attack, Malware communication, Reconnaissance activities, Exploits, Unauthorized access attempts, Data breaches are some examples of cyber threats that are still sophisticated enough and frequent that it poses a great risk to cloud service providers and users [7], [13]. Cloud infrastructures have been protected using traditional security approaches such as firewall, signature-based IDS and rule-based monitoring. These techniques work well when the attacks are known, but they have been found to be less effective against previously unknown attacks and zero-day attacks, due to their reliance on predefined signatures and manually written rules [6], [22]. This has led to an increasing need for intelligent intrusion detection systems with the ability to learn complex network behaviours and adapt to new attack patterns automatically. The potential of machine learning to analyze large volumes of network traffic and accurately and adaptively detect malicious activities while identifying legitimate traffic has proven to be a promising solution to this challenge [7, 15]. Various machine learning models have been utilized for intrusion detection, such as Decision Trees, Support Vector Machines, Random Forest, Gradient Boosting models, and Deep Learning models [8, 9, 16]. Of these techniques, ensemble learning has proven to be quite popular for its ability to produce a strong learner from a number of weak learners. Introduced by Chen and

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Guestrin, XGBoost (Extreme Gradient Boosting) is one of the most powerful and popular ensemble learning algorithms due to its scalability, computational efficiency and performance of providing superior predictive power in various classification tasks [3]. XGBoost uses gradient boosting methods and regularization techniques to solve the problem of prediction error iteratively and suppress overfitting, which is very suitable for cybersecurity applications like anomaly detection and intrusion detection [3, 14]. While machine learning models can have impressive predictive power, many of these systems are black box in nature, providing little insight as to how decisions are made. In such cybersecurity contexts, it is a critical problem as security analysts and administrators need to be provided with explanations of attack classifications before taking corrective measures [4, 20]. A system which marks network traffic as malicious without giving reasons for the decision can affect user trust and make it difficult to investigate threats provided.A system that flags network traffic as malicious and offers no explanation of what has caused the decision may diminish user confidence and make effective threat investigation difficult to accomplish. Therefore, the research fields of explainability and interpretability are becoming more relevant in the field of artificial intelligence and cyber security [5, 21]. Explainable Artificial Intelligence (XAI) is a step forward by offering methods to understand the "how" and "why" behind data predictions made by machines. SHAP (SHapley Additive exPlanations) is one of the most reliable and theoretically-sound methods for model interpretation among the many XAI approaches [4]. SHAP is derived from cooperative game theory and assigns contribution values to features, allowing analysts to interpret the contribution of a specific feature of the network to the prediction results [4, 5]. SHAP can be used to improve the transparency, accountability and trust of machine learning models for intrusion detection systems by identifying the most influential features associated with malicious activity. In recent years, researchers have been emphasizing the significance of explainability techniques in cybersecurity applications, where they can help in making informed decisions and increase the trust in automated security solutions [21]. While there are many studies that have explored machine learning and deep learning methods for intrusion detection, existing methods have paid more attention to detecting intrusions with the highest accuracy rather than giving much attention to explainability and transparency [8, 11, 16]. Moreover, some deep learning structures require a lot of computing power and tend to generate hierarchical models that are hard to understand. This leaves room for research in the development of efficient, accurate and explainable intrusion detection systems that are applicable to real-world cloud security systems [11], [21]. The proposed framework is designed to detect malicious network activities like DDoS attacks, exploits, reconnaissance attempts, fuzzing attacks, communication activities associated with malware, and unauthorized access attempts in the cloud infrastructure. It applies the UNSW-NB15 dataset which is a contemporary benchmark dataset created to address the drawbacks of existing intrusion detection datasets and offer realistic network traffic cases for cybersecurity research [1, 2]. The features of the network traffic are pre-processed and encoded in a machine-readable format prior to their being fed into the XGBoost attack detection classifier. Then, SHAP is used to interpret the model predictions and to determine the most significant features that make up the decisions for intrusion classification [4, 5]. The main goal of this research is to design an intrusion detection framework that will be both high predictive and transparent and interpretable. The proposed approach is evaluated experimentally on UNSW-NB15 which shows that the proposed approach achieves an overall accuracy of 95.13% with good precision, recall and F1-score. The results show that the proposed framework not only boosted the performance in intrusion detection but also made the decision making process more interpretable, making it a realistic and reliable framework for contemporary cloud security applications [3, 4, 21]. The proposed framework makes it a valuable addition to the realm of intelligent and explainable cybersecurity systems, with its blend of prediction accuracy, computational efficiency, and explainability.

2. Literature Review

In recent developments, an increasing amount of work has been done to embed machine-learning methods to enhance the identification of malicious network activities in the cloud. Satisfying these limitations of these earlier benchmark datasets, Moustafa and Slay created the UNSW-NB15 dataset, which comes with realistic network traffic for intrusion detection research [1]. XGBoost is a scalable gradient boosting framework that

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can achieve high predictive performance in classification problems, proposed by Chen and Guestrin [3]. SHAP was a novel unified interpretable approach proposed by Lundberg and Lee that allows us to quantify the contribution of features in order to interpret the prediction of a machine learning model [4]. In a comprehensive survey, Buczak et al. described different machine learning techniques used for the security of networks, showing how data-driven intrusion detection methods work well [7]. Shone et al. suggested a deep learning based intrusion detection framework whose result was more accurate in attack classification than conventional approach [9]. Ferrag et al. investigated the uses of deep learning algorithms in cybersecurity, highlighting that precise and adaptive intrusion detection system for contemporary networks is crucial [11]. Various intrusion detection techniques have been studied thus far, and Khraisat et al. discussed the issues faced with data quality, false alarm rate, and attack variation [13]. Aljawarneh et al. proposed an intrusion detection model based on a hybrid technique of feature selection and machine learning for better classification [14] Ahmad and colleagues performed a systematic survey of the use of machine learning and deep learning technologies in network intrusions and stated their suitability and increasing importance in the field of intelligence security systems [16]. Recently, Bhati et al. discussed the role of XAI in addressing intrusive tasks such as intrusion detection which requires transparency and credibility in the model [21]. Previous research has led to some successful detection results but most are oriented toward accuracy with little interpretability. Hence, by combining XGBoost with SHAP, an efficient attack detection mechanism to boost transparency and trust in cloud security decision-making processes becomes feasible [3], [4], [21].

3. Methodology

3.1 Study Design

This study introduces an Explainable XGBoost-Based Intrusion Detection Framework to improve the security of the cloud by accurately detecting malicious network activities. It fuses the machine learning-based attack detection with explainability using SHAP to attain both prediction and interpretability. The methodology involves data collection, data preprocessing, feature encoding, training the model, making predictions, assessing the model, and analyzing the model's explainability.

3.2 Study Population

The study is based on the UNSW-NB15 dataset which consists of modern network traffic records both from legitimate and malicious traffic. After pre-processing, 257,673 instances of network traffic and 42 predictive features are provided.

The attack categories include:

- Generic
- Exploits
- Fuzzers
- DoS
- Reconnaissance
- Analysis
- Backdoor
- Shellcode
- Worms

The target variable is represented as:

$$y = \begin{cases} 0, & \text{Normal Traffic} \\ 1, & \text{Malicious Traffic} \end{cases}$$

where:

- $y = 0$indicates legitimate network traffic.
- $y = 1$indicates malicious network traffic.

3.3 Data Preprocessing

Label Encoding is used to convert categorical attributes like protocol, service type, connection state to numerical values.

The transformation is represented as:

$$X_{encoded} = LabelEncoder(X_{categorical})$$

Role

The encoding process is a mapping of the attributes of a network in text into numerical values which are readable by the machine.

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Contribution

This allows the XGBoost algorithm to learn effectively from categorical network features when training the model.

3.4 Feature Matrix Construction

The feature matrix is defined as:

$$X = \{x_1, x_2, x_3, \dots, x_n\}$$

where:

- x_1= duration
- x_2= protocol
- x_3= service
- ...
- x_n= network traffic features

Target variable:

$$Y = \{0,1\}$$

Role

All network properties used to detect attacks are contained in the feature matrix.

Contribution

It gives input data needed for learning traffic behavior patterns.

3.5 XGBoost Classification Model

Extreme Gradient Boosting (XGBoost) algorithm is used in the proposed framework for intrusion detection.

The prediction function is:

$$\hat{y} = \sum_{k=1}^{K} f_k\,(x_i)$$

where:

- $\hat{y}$= predicted class
- K= number of decision trees
- f_k= individual tree model
- x_i= network traffic sample

Role

XGBoost is made up of several weak learners that are combined into a strong predictive model.

Contribution

It enhances classification accuracy as well as minimising prediction error by means of iterative learning.

3.6 Objective Function

The XGBoost objective function is expressed as:

$$Obj = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where:

- L= loss function
- y_i= actual value
- $\hat{y}_i$= predicted value
- Ω= regularization term

Role

The objective function is designed to minimize the error of classification with a control of the complexity of the model.

Contribution

It helps to avoid overfitting and enhances the generalization ability.

3.7 Gradient Boosting Update

Each new tree will try to fix the errors that were made in the previous one.

$$F_m(x) = F_{m-1}(x) + \gamma_m h_m(x)$$

where:

- $F_m(x)$= updated prediction
- $F_{m-1}(x)$= previous prediction
- $h_m(x)$= newly generated tree
- γ_m= learning rate

Role

The model learns from previous mistakes.

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Contribution

This optimization is done in an iterative way and enhances the attack detection performance.

3.8 Accuracy Evaluation

The accuracy is the percentage of correctly classified traffic records.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100$$

where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Role

Checks the correctness of the whole model.

Contribution

The proposed model obtained:

$$Accuracy = 95.13\%$$

3.9 Precision Evaluation

Precision measures attack prediction reliability.

$$Precision = \frac{TP}{TP + FP}$$

Role

Describes the percentage of the attacks that are attacks.

Contribution

Minimises cloud security false alarms.

3.10 Recall Evaluation

Recall: an attack on detection capability.

$$Recall = \frac{TP}{TP + FN}$$

Role

Calculates the percentage of successful attacks that are detected.

Contribution

Improves identification of malicious activities.

3.11 F1-Score Evaluation

The harmonic mean of Precision and Recall is equal to:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Role

Offers a reasonable measure of evaluation.

Contribution

Provinces consistent detection of intrusions across various distributions of attacks.

3.12 SHAP Explainability Analysis

Each feature's SHAP is calculated as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

where:

- ϕ_i= SHAP value
- S= subset of features
- N= complete feature set

Role

Calculates the contribution that each feature makes to the prediction.

Contribution

Gives transparency by giving details of why network traffic is considered malicious or normal.

3.13 Statistical Analysis

The following is used to statistically assess the model performance:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- ROC Curve
- Feature Importance Analysis
- SHAP Explainability

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The proposed Explainable XGBoost framework successfully achieved an overall classification accuracy of 95.13%, which is useful for detecting malicious traffic on the cloud. This predictive performance and explainability make it a good fit for real-world cloud security use cases.

4. Results

4.1 Experimental Results

The proposed Explainable XGBoost-Based Intrusion Detection Framework was tested with the UNSW-NB15 dataset with the goal of distinguishing between network traffic of normal and malicious. The model performed well with an accuracy of 95.13%, highlighting its effectiveness in identifying cyber threats in cloud environments. The usage of SHAP explainability further improved the understanding of the model by determining the most important network features affecting the prediction of attacks.

4.2 Performance Analysis

Standard evaluation metrics: Accuracy, Precision, Recall, and F1-Score were used to evaluate the performance of the proposed model. The results show that the XGBoost classifier effectively separated the malicious traffic and normal traffic with high accuracy.

Table 1. Performance Metrics of the Proposed Model

Metric	Value (%)
Accuracy	95.13
Precision	97.00
Recall	96.00
F1-Score	96.00

The precision value is high: this means that the model doesn't produce too many false alarms, and the recall value is good: this means that the model catches most of the malicious events. F1-score is a measure of the balanced precision and recall.

4.3 Dataset Summary

Table 2. Dataset Statistics

Parameter	Value
Total Samples	257,673
Total Features	42
Attack Records	164,673
Normal Records	93,000

A wide range of normal and attack traffic is recorded in the dataset which allows for a thorough assessment of the performance of intrusion detection systems.

4.4 Important Feature Analysis

Table 3. Top Influential Features Identified by XGBoost

Rank	Feature
1	sttl
2	ct_state_ttl
3	proto
4	sload
5	dload
6	ct_srv_src
7	service
8	state
9	ct_dst_src_ltm
10	rate

The results of the feature importance analysis showed that traffic related features like source time-to-live, protocol types, traffic load metrics and connection statistics are important features for distinguishing malicious network behavior from legitimate traffic.

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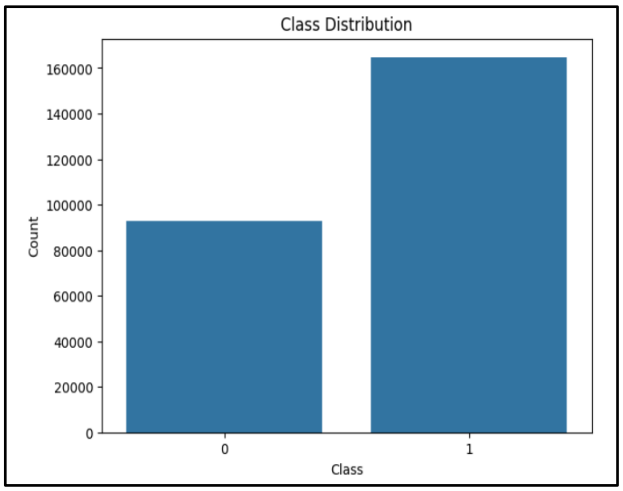
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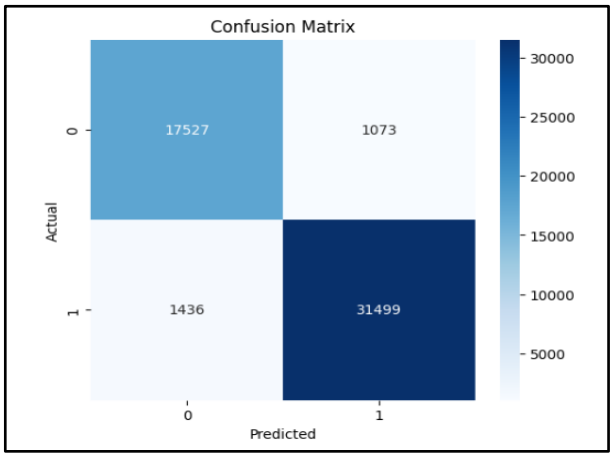
4.5 Figure Analysis

Figure 1. Class Distribution of Network Traffic



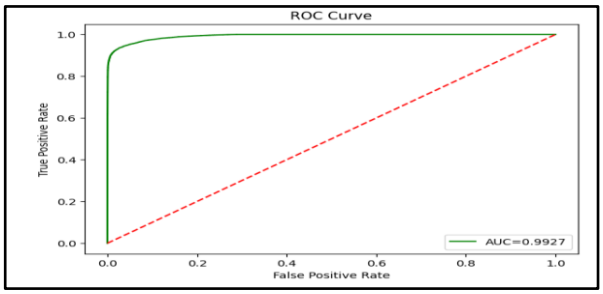
The class distribution graph shows the percentage of attack and normal records in the data. The visualization also shows that the dataset has a large number of attack instances, which allows for the effective training and evaluation of intrusion detection models.

Figure 2. Confusion Matrix of the Proposed XGBoost Model



The confusion matrix shows the results of the classification of normal and malicious traffic. It is very clear from the figure that the number of samples misclassified is relatively small and the number of samples positively detected is also relatively small, the detection capability is good.

Figure 3. ROC Curve of the Proposed Model

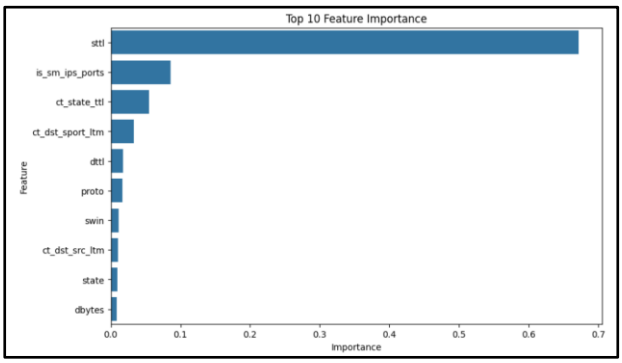


The ROC curve assesses the model discrimination over various threshold values. The curve stays near the upper left corner indicating a very good separation of attack and normal traffic and showing the strength of the classifier.

Table 4: Top 10 Feature Importance Ranking

Rank	Feature Number	Feature Name	Feature Importance
1	9	sttl	0.671112
2	41	is_sm_ips_ports	0.086205
3	31	ct_state_ttl	0.054556
4	34	ct_dst_sport_ltm	0.033446
5	10	dttl	0.017714
6	1	proto	0.016746
7	19	swin	0.011048
8	35	ct_dst_src_ltm	0.010309
9	3	state	0.009291
10	7	dbytes	0.008519

Figure 4. Top 10 Feature Importance Scores



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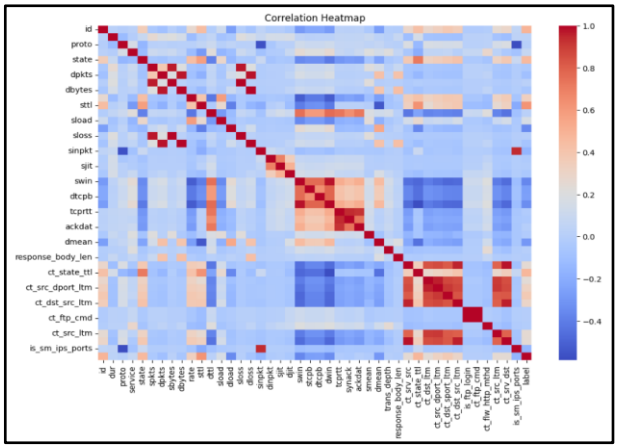
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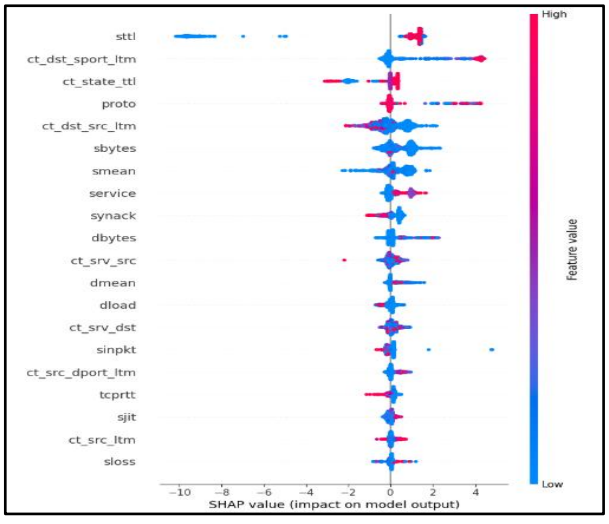
This number indicates the most important features that were used in the classification by the XGBoost model. The protocol behavior, packet attributes and connection statistics play a major role in intrusion detection performance.

Figure 5. Correlation Heatmap of Network Features



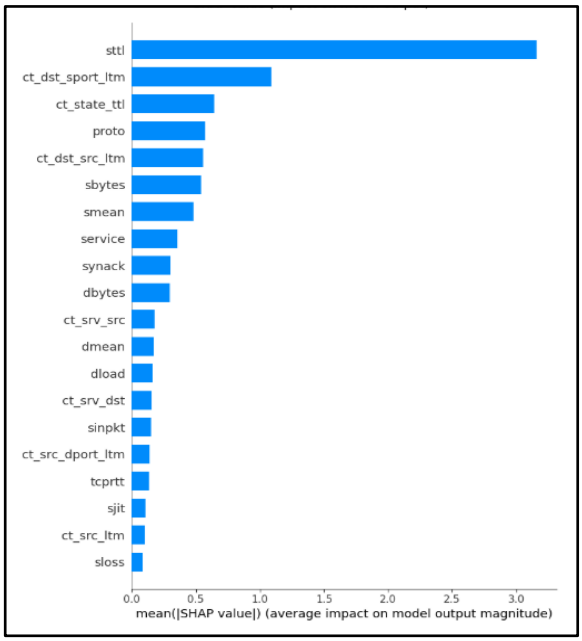
The correlation heatmap depicts the interconnections between the features of the dataset. The figure helps to understand the dependencies of features and helps to see how the traffic was generated when training the model.

Figure 6. SHAP Summary Plot



This summary plot (SHAP) describes the impact of the features on model predictions. It shows how various feature values affect classification decisions on attack or not and can offer transparency to intrusion detection process.

Figure 7. SHAP Feature Importance Bar Plot



SHAP bar plot ranks the features by their average effect on predictions of the model. The visualization also highlights the importance of some of the features related to the traffic and helps in the explanation of the proposed explainable intrusion detection system.

4.6 Overall Findings

The experimental results validate the effectiveness of the proposed Explainable XGBoost Based Intrusion Detection Framework to detect cloud network intrusions. The model has an accuracy of 95.13% with high value of precision, recall, and F1 scores. Also, SHAP explainability helped security analysts to gain insights into the decision-making process of the classifier, offering meaningful insights into the identified attacks. The results validate the proposed approach with high detection performance and interpretability, which are desirable in contemporary cloud security systems.

5. Discussion

5.1 Interpretation and Contextualization of Findings

The experimental results verified that the proposed Intrusion Detection Framework based on Explainable XGBoost was able to detect malicious activities in the network within cloud settings. The overall accuracy of the model was 95.13%, which indicated high classification ability and high precision, recall and F1

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values for the model. The result is in line with existing research on how well ensemble machine learning techniques can be used for intrusion detection. The value of this framework is enhanced when SHAP explainability is added, as it offers light on what matters for making a classification. Feature importance analysis showed that characteristics of network traffic including the protocol behaviour, packet transmission patterns and connection statistics play an important role in the process of attack detection. The proposed framework has the advantages of predictability and explainability compared to the traditional black-box intrusion detection models, which are more suitable for specific cloud security applications.

5.2 Significance, Implications and Limitations

The recommended model will have a major impact on the current cloud security systems because of its capability to identify cyber threats at an early stage and to give transparency on decision making. This model is explainable, which helps security administrators to grasp the nature of the attack, enhance trust in automated security solutions, and shorten response times in case of an attack. The performance obtained illustrates the capability of the proposed ensemble approach XGBoost + SHAP for practical intrusion detection anomalies. There are some limitations of this study. The framework has been tested against a benchmark dataset and could not be tested with real cloud traffic as attack types may change. Furthermore, supervised learning is the central issue in the model and it performs well when the training data is of labeled type. These deficiencies can be addressed by future research integrating the real-time traffic propagation from the cloud, adapting the controls to learning mechanisms and incorporating advanced deep learning methods to boost the detection accuracy and robustness to newly arising cyber threats.

6. Conclusion

This is to cope with the increasing threat to the detection of malicious activities in cloud computing environments. The framework adopted used the UNSW-NB15 dataset and used XGBoost as the main classification algorithm to classify network traffic as normal or malicious. To improve transparency and interpretability, SHAP analysis is added into the framework, so that features most impacting intrusion detection decision can be identified. Experimental results showed that the proposed model has an overall accuracy of 95.13%, precision, recall, and F1-score values are high, thus proving its usefulness in detection of cyber threats like DoS attack, Exploit, Reconnaissance Attack and other threats. It was then found that explainability techniques can further increase user trust and understanding of a security system using machine learning by offering relevant insights for the model's predictions. The proposed framework is a viable solution, since the intrusion detection is accurate and interpretable decision-making occurs in the cloud security application. In addition, it emphasizes the possibility of harmonizing XAI with powerful ML algorithms aimed at improving the security operations. Future research could address real-time cloud traffic analysis, adaptive learning approaches, and hybrid deep learning architectures, among other related focuses, to further enhance the accuracy of detection and the resistance to new emerging cyber threats.

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